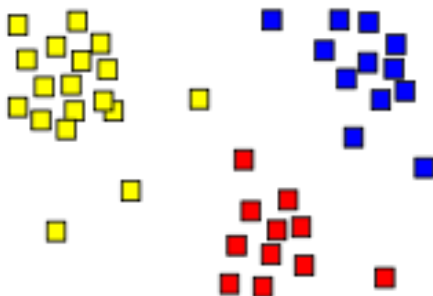


Bayesian model-based clustering using finite mixture models

Gertraud Malsiner-Walli

Clustering

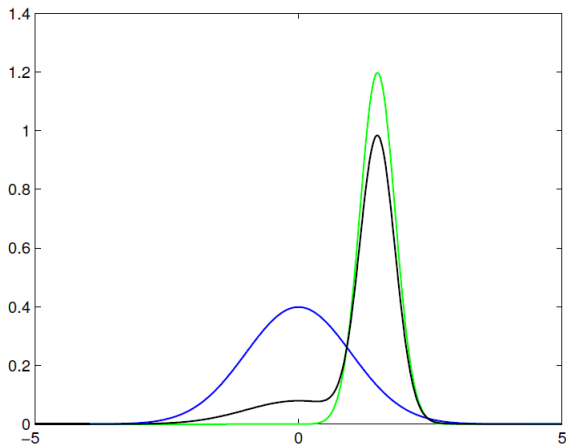
- Cluster analysis aims at partitioning a set of objects into groups such that objects within the same group (called a cluster) exhibit greater similarity to one another (in some specific sense defined by the analyst) than to those in other groups (clusters).
- Idea of **model-based clustering**: use a statistical model (finite mixture distribution) where each cluster is generated by its own density.



Finite mixture models

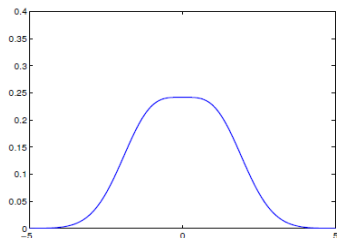
- Finite mixture models are a flexible model class:
 - ▶ They allow modeling unobserved heterogeneity / finding groups in data
 - ⇒ useful for model-based clustering
 - ▶ They are a semi-parametric tool to estimate arbitrary distribution functions
- Applications in
 - ▶ astronomy
 - ▶ biology
 - ▶ economics
 - ▶ ...

Mixture of 2 normals

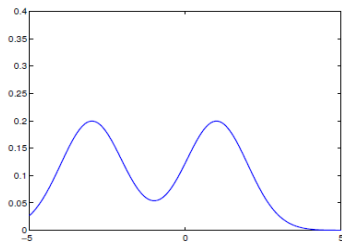


Mixture $y \sim 0.2 \mathcal{N}(0, 1) + 0.8 \mathcal{N}(1.5, 1/9)$

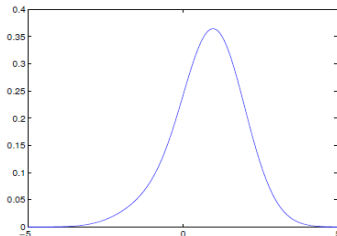
Mixtures with different means and weights



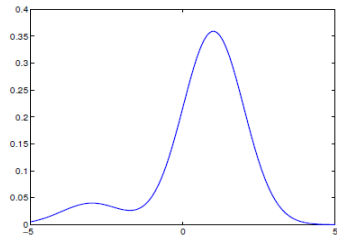
$$y \sim 0.5\mathcal{N}(-1, 1) + 0.5\mathcal{N}(1, 1)$$



$$y \sim 0.5\mathcal{N}(-3, 1) + 0.5\mathcal{N}(1, 1)$$



$$y \sim 0.1\mathcal{N}(-1, 1) + 0.9\mathcal{N}(1, 1)$$



$$y \sim 0.1\mathcal{N}(-3, 1) + 0.9\mathcal{N}(1, 1)$$

Modelling heterogeneity

- Population made up of K subgroups with relative group sizes $\eta_1, \dots, \eta_K$.
- Random variable Y , which is heterogeneous across the subgroups, but homogeneous within the subgroups:
 - ▶ Y has a different probability distribution $p_k(y)$ in each group, arising from the same parametric family but with different parameters θ_k :

$$p_k(y|\theta_k) = p(y|\theta_k)$$

- We record for each observation:
 - ▶ random variable Y
 - ▶ group indicator $S \in \{1, \dots, K\}$
 - ⇒ The joint probability density $p(y, S)$ is given by

$$p(y, S) = p(y|S)p(S) = p(y|S)\eta_S$$

Mixtures with observed group indicator

- A finite mixture arises, if it is not possible to record the group indicator S : we can only observe the random variable Y .

The marginal density is given by the mixture density:

$$p(y) = \sum_{k=1}^K p(y, S) = \sum_{k=1}^K \eta_k p(y|\theta_k)$$

where $\eta_1, \dots, \eta_K$ are called 'mixing proportions' or 'weights', with

$$\sum_{k=1}^K \eta_k = 1 \quad \text{and} \quad \eta_k > 0 \quad \forall k$$

Statistical inference on mixture models

$$p(y) = \sum_{k=1}^K p(y, S) = \sum_{k=1}^K \eta_k p(y|\theta_k)$$

For observations $y_1, \dots, y_N$ drawn randomly from a finite mixture distribution of K groups, three kinds of statistical inference problems have to be considered:

1. Estimation of the component parameters and weights
 $\vartheta = (\theta_1, \dots, \theta_K, \eta_1, \dots, \eta_K)$.
2. Allocation of the observations to the groups through $\mathbf{S} = (S_1, \dots, S_N)$.
3. Specification of the number of groups K .

Case 1: Bayes Inference when group indicators are known

- **Complete data likelihood:**

$$\begin{aligned} p(\mathbf{y}, \mathbf{S} | \boldsymbol{\theta}, \boldsymbol{\eta}) &= \prod_{i=1}^N \prod_{k=1}^K \left(p(y_i | \theta_k) \eta_k \right)^{I_{\{S_i=k\}}} \\ &= \prod_{k=1}^K \left(\prod_{i: S_i=k} p(y_i | \theta_k) \right) \left(\prod_{k=1}^K \eta_k^{N_k(\mathbf{S})} \right) \end{aligned}$$

where $N_k(\mathbf{S}) = \#\{S_i = k\}$ is the number of observations in component k

- **Prior:**

$$p(\theta_1, \dots, \theta_K, \boldsymbol{\eta}) = \prod_{k=1}^K p(\theta_k) p(\boldsymbol{\eta})$$

where $\boldsymbol{\eta} = (\eta_1, \dots, \eta_K)$.

- ▶ Prior on θ_k depends on the parameter.
- ▶ Prior on $\boldsymbol{\eta}$: Dirichlet distribution $\mathcal{Dir}(\gamma_{0,1}, \dots, \gamma_{0,K})$.

Excursion: Dirichlet distribution

Definition

$\mathbf{X} = (X_1, \dots, X_K)$ follows a **Dirichlet distribution** $\text{Dir}(\alpha_1, \dots, \alpha_K)$ with parameters $\alpha_1, \dots, \alpha_K$, if its pdf is given as

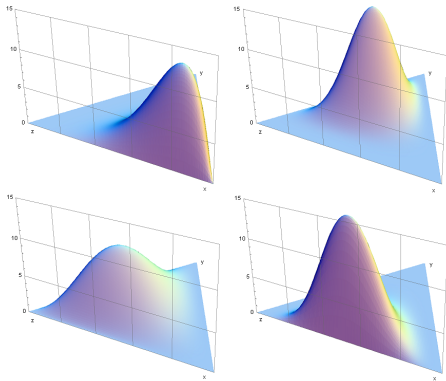
$$f_{\mathbf{X}}(\mathbf{x}) = \frac{1}{B(\alpha)} x_1^{\alpha_1-1} \cdots x_K^{\alpha_K-1},$$

with $B(\alpha) = \frac{\prod_{k=1}^K \Gamma(\alpha_k)}{\Gamma(\sum_{k=1}^K \alpha_k)}$, $\sum_{k=1}^K \alpha_k = 1$.

- Positive pdf only for $\sum_{k=1}^K x_k = 1, x_k > 0$,
i.e. the $K - 1$ -dimensional simplex E_K
- Special cases:
 - ▶ Beta distribution for $K = 2$: $X \sim \mathcal{B}(\alpha_1, \alpha_2)$
 - ▶ $\alpha_1 = \dots = \alpha_K = 1$: uniform distribution on E_K .

Dirichlet distribution

Dirichlet distribution for $(x_1, x_2, x_3) \sim \text{Dir}(\alpha_1, \alpha_2, \alpha_3)$, for different values of α_k .



Bayes Inference for mixtures with known group indicators

Posterior:

$$p(\boldsymbol{\theta}, \boldsymbol{\eta} | \mathbf{y}, \mathbf{S}) \propto \prod_{k=1}^K \left(\prod_{i: S_i=k} p(y_i | \boldsymbol{\theta}_k) p(\boldsymbol{\theta}_k) \right) \left(\prod_{k=1}^K \eta_k^{N_k(\mathbf{S})} p(\boldsymbol{\eta}) \right)$$

Bayesian estimation can be carried out independently for each component parameter $\boldsymbol{\theta}_k$ and the weights $\boldsymbol{\eta}$:

- Inference for $\boldsymbol{\theta}_k$ as usual for the parameter in group k
e.g. $y_i | S_i \sim \mathcal{N}(\mu_k, 1) \Rightarrow \theta_k = \mu_k$
- Inference for $\boldsymbol{\eta}$:
 $\boldsymbol{\eta} \sim \text{Dir}(\gamma_{0,1}, \dots, \gamma_{0,K}) \Rightarrow \boldsymbol{\eta} | \mathbf{S} \sim \text{Dir}(\gamma_1(\mathbf{S}), \dots, \gamma_K(\mathbf{S}))$ with
 $\gamma_k(\mathbf{S}) = \gamma_{0,k} + N_k(\mathbf{S}).$

Finite Mixture models

- Typically the group indicators (allocations of subjects to groups) are not observed
- The marginal density of y is then a **finite mixture** distribution of K components

$$p(y) = \sum_{S=1}^K \eta_k p(y|\theta_k)$$

- Mixture Likelihood:

$$\prod_{i=1}^N \left(\sum_{k=1}^K \eta_k p(y_i|\theta_k) \right)$$

- No conjugate prior exists!
 - $\implies$ the posterior $p(\theta, \eta|\mathbf{y})$ cannot be derived analytically
 - $\implies$ **Data augmentation** with unknown allocations **S** as latent observations.

Parameter estimation through data augmentation and MCMC

⇒ Practical Bayesian estimation follows Dempster and Rubin (1977) considering the finite mixture model as a missing data problem which can be completed and simplified by introducing the allocations $\mathbf{S}$ as **latent variables**.

⇒ The benefits of this data augmentation are that

- conditionally on knowing $\mathbf{S}$, we are back in the conjugate setting of the complete data Bayesian estimation,
- conditionally on knowing the parameters ϑ , we are back to the classification problem, where the posterior distribution of the allocations takes a very simple form.

⇒ It is rather straightforward to sample from the posterior using MCMC methods, in particular Gibbs sampling.

Bayes Inference for unknown group indicators

Posterior of $(\mathbf{S}, \boldsymbol{\theta}, \boldsymbol{\eta})$:

Gibbs Sampling: Start with some classification $\mathbf{S}^{(0)}$ and repeat the following steps

- 1 Sample the parameters $(\boldsymbol{\theta}, \boldsymbol{\eta})$ conditional on the classification $\mathbf{S}^{(m-1)}$ (as for known allocations)
- 2 Classification of each observation y_i conditional on knowing $(\boldsymbol{\theta}, \boldsymbol{\eta})$: sample $\mathbf{S}_i$ independently for $i = 1, \dots, n$ from the conditional posterior distribution

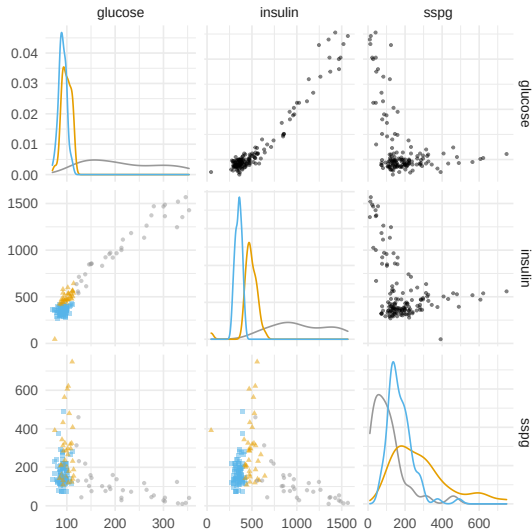
$$P(\mathbf{S}_i | (\boldsymbol{\theta}, \boldsymbol{\eta})^{(m)}, y_i) \propto p(y_i | \boldsymbol{\theta}_k^{(m)}) \boldsymbol{\eta}_k^{(m)}.$$

K is known

Example: Diabetes data I

- $N = 145$ adults with three types of diabetes ('normal', 'overt', 'chemical').
- Three measurements are available:
 - ▶ `glucose`: the area under the plasma glucose curve after a three hour oral glucose tolerance test (OGTT),
 - ▶ `insulin`: the area under the plasma insulin curve after a three hour OGTT,
 - ▶ `sspg`: the steady state plasma glucose level.
- Aim of the analysis: recover the three groups.

Example: Diabetes data II



Model and priors

$$\text{Sampling model: } \mathbf{y} \sim \sum_{k=1}^K \eta_k f_{\mathcal{N}}(\mu_k, \Sigma_k)$$

$$\text{Priors: } \mu_k \sim \mathcal{N}(\mathbf{b}_0, \mathbf{B}_0), k = 1, \dots, K,$$

$$\Sigma_k \sim \mathcal{W}^{-1}(c_0, \mathbf{C}_0), k = 1, \dots, K,$$

$$\mathbf{C}_0 \sim \mathcal{W}(g_0, \mathbf{G}_0),$$

$$\eta_K \sim \text{Dir}(\gamma)$$

MCMC scheme

Given some initial parameter values η_K and Θ_K , iterate Steps 1 and 2:

- 1 Classification of each observation $\mathbf{y}_i$, $i = 1, \dots, N$, conditional on η_K and Θ_K :

- 1 For $i = 1, \dots, N$: sample S_i from

$$P(S_i = k | \mathbf{y}_i, \vartheta_K) \propto \eta_k f(\mathbf{y}_i | \theta_k).$$

- 2 Parameter simulation conditional on the classification $\mathbf{S} = (S_1, \dots, S_N)$:

- 1 Sample the mixture weights η_K from

$$\eta_K | \mathbf{S} \sim \mathcal{D}(\gamma_1, \dots, \gamma_K),$$

where

$$\gamma_k = \gamma + N_k$$

with N_k is the number of observations assigned to component k .

- 2 Sample the component-specific parameters θ_k , $k = 1, \dots, K$ conditional on $\mathbf{y}$ and $\mathbf{S}$ from their conditional conjugate posteriors.
- 3 Sample hyper-parameters ϕ conditional on Θ_K .

Sampling steps 2 and 3

To fit the finite mixture of multivariate Gaussian distributions to the `diabetes` data set, Steps 2(b) and 2(c) are specifically given by:

2(b) For all $k = 1, \dots, K$, sample

$$\begin{aligned}\boldsymbol{\mu}_k | \mathbf{y}, \mathbf{S}, \boldsymbol{\Sigma}_k &\sim \mathcal{N}(\mathbf{b}_k, \mathbf{B}_k), \\ \boldsymbol{\Sigma}_k | \mathbf{y}, \mathbf{S}, \boldsymbol{\mu}_k &\sim \mathcal{W}^{-1}(c_k, \mathbf{C}_k),\end{aligned}$$

with

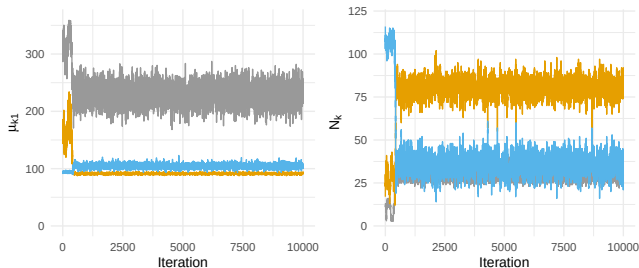
$$\begin{aligned}\mathbf{b}_k &= \mathbf{B}_k(\mathbf{B}_0^{-1}\mathbf{b}_0 + \boldsymbol{\Sigma}_k^{-1}N_k\bar{\mathbf{y}}_k), & \mathbf{B}_k &= (\mathbf{B}_0^{-1} + N_k\boldsymbol{\Sigma}_k^{-1})^{-1}, \\ c_k &= c_0 + \frac{N_k}{2}, & \mathbf{C}_k &= \mathbf{C}_0 + \frac{1}{2} \sum_{i:S_i=k} (\mathbf{y}_i - \boldsymbol{\mu}_k)(\mathbf{y}_i - \boldsymbol{\mu}_k)^\top,\end{aligned}$$

where $\bar{\mathbf{y}}_k = \frac{1}{N_k} \sum_{i:S_i=k} \mathbf{y}_i$.

2(c) Sample

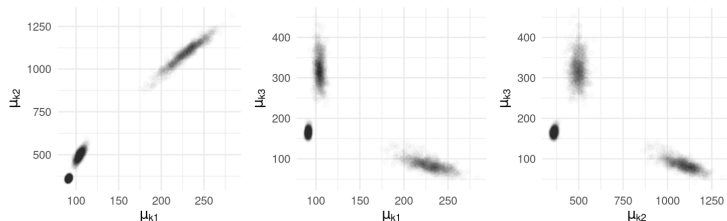
$$\mathbf{C}_0 | (\boldsymbol{\Sigma}_k)_{k=1, \dots, K} \sim \mathcal{W}(g_0 + Kc_0, \mathbf{G}_0 + \sum_{k=1}^K \boldsymbol{\Sigma}_k^{-1}).$$

Diagnostic plots



Trace plots of the μ_{k1} draws of the first variable (left) and the number of observations N_k allocated to components (right), in different colors for the three components.

Scatter plots of the sampled component means



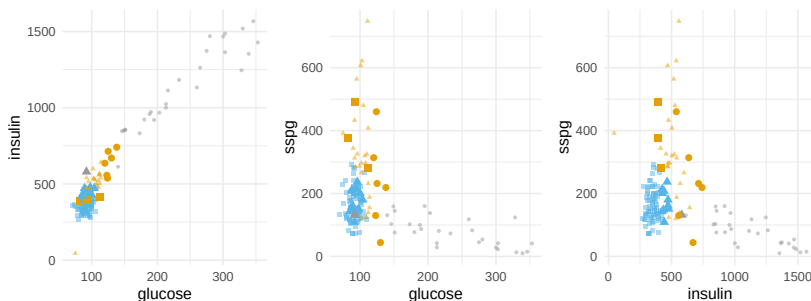
Bivariate scatter plots of the μ draws for all three components and for the three variables

Final partition

- The final partition of the data is obtained by relabeling the assignment vectors $\mathbf{S}$ according to the unique labeling obtained in the ppr approach and then assigning each observation to the component to which it was assigned most frequently during MCMC sampling.
- Confusion matrix:

True	Estimated		
	1	2	3
Overt	27	6	0
Chemical	1	24	11
Normal	0	3	73

Final clustering



Bivariate scatter plots of the clinical measurements, with colors indicating the component membership estimated using the posterior modes of the component assignments. The size of misclassified observations is increased.

K is unknown

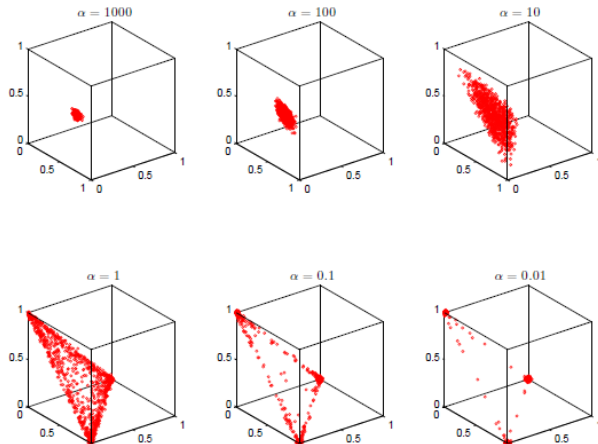
- Issue: How many clusters are in the data?
- Model selection problem: fit finite mixtures with a different number of components K and compare the models by the marginal likelihood (difficult to compute).
- Solution 1: Sparse finite mixture approach (Malsiner-Walli, Frühwirth-Schnatter, and Grün, 2016)
- Solution 2: Mixture with a prior on K (Frühwirth-Schnatter, Malsiner-Walli, and Grün, 2021)

Overfitting mixtures

- The usual assumption is that each component captures exactly one data cluster.
 - What happens if there are more components K than clusters K_+ ?
- ⇒ Superfluous components can be left empty or filled components are split into two or more components.
- The behavior of the posterior distribution to follow the first or second strategy depends on the Dirichlet prior on the weights.
 - If the Dirichlet parameter γ is very small, superfluous components become empty during MCMC. Otherwise, components are split (Rousseau and Mengersen, 2011)

Dirichlet prior

Samples $\eta \sim \text{Dirichlet}(\alpha, \alpha, \alpha) \propto \prod_{k=1}^3 \eta_k^{\alpha-1}$, $\sum_{k=1}^3 \eta_k = 1$



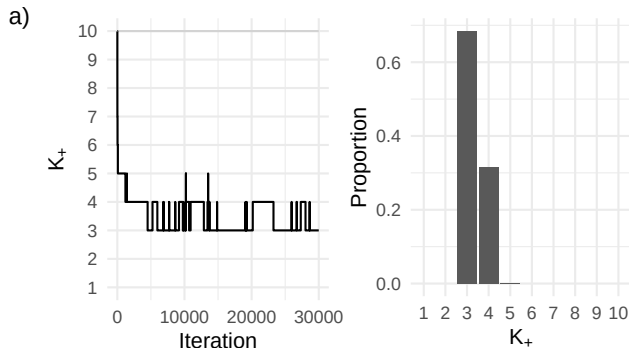
Plot is taken from the slides "Investigations in variable selection for Bayesian mixture models", Chris Holmes and Chris Yau, Edinburgh, 2010.

Sparse finite mixture

Idea:

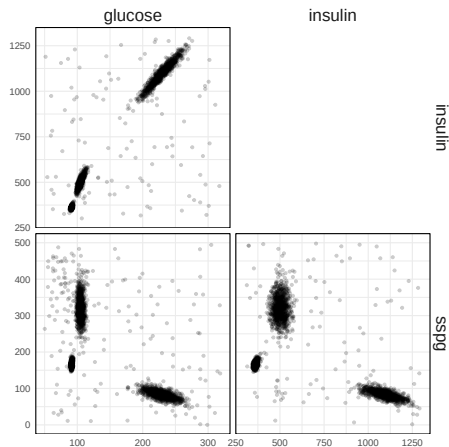
- Specify an overfitting mixture where K is much larger than the number of clusters K_+ in the data.
- Specify a Dirichlet prior on the weights with a very small parameter γ .
- Then, during MCMC sampling redundant component should be emptied.
- Estimate the number of clusters $\hat{K}_+$ by the mode of the posterior of K_+ .
- Take only draws from iterations where $K_+ = \hat{K}_+$. Discard the draws from empty components.

Sparse finite mixture with $K = 10$, $\gamma = 0.01$



Trace plot of the filled components and posterior of K_+ .

Sparse finite mixture: Scatter plot of the sampled component means



The draws from filled components are overlaid by a diffuse cloud of draws from empty components.

Mixture of finite mixtures

- In Bayesian analysis a natural approach to account for the unknown number of components K is to introduce a prior on K ,

$$K \sim p(K),$$

and make inference on the posterior of K and K_+
 $\implies$ 'mixture of finite mixtures (MFM) model'

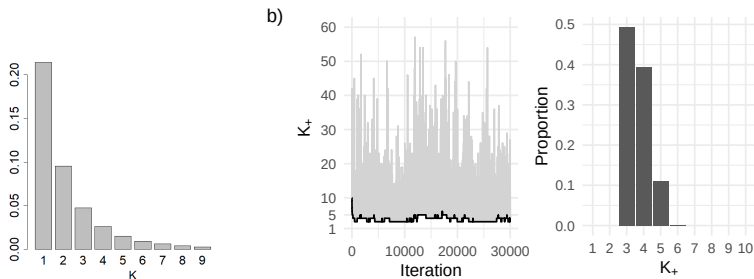
- Issue: For inference a trans-dimensional sampler is required (e.g., reversible jump MCMC is difficult to tune).
- Idea: Telescoping sampler: extends the usual MCMC scheme by including also a step where K is explicitly sampled on the current partition of the data and a step to obtain parameters for empty components:

3. Sample K conditional on current cluster sizes $(N_1, \dots, N_{K_+})$ and γ_K from

$$p(K|(N_1, \dots, N_{K_+}), \gamma_K) \propto \frac{K!}{(K-K_+)!} \frac{\Gamma(K\gamma_K)}{\Gamma(K\gamma_K+N)} \prod_{k=1}^{K_+} \frac{\Gamma(N_k+\gamma_K)}{\Gamma(1+\gamma_K)} p(K).$$

4. Add $K - K_+$ empty components with component-specific parameters sampled from the priors.

MFM: Diabetes data



- Dynamic MFM with $K - 1 \sim \text{BNB}(1, 4, 3)$ and $\gamma = 0.5/K$.
- Left: Prior on K .
- Middle and right: Trace plot of the number of components (gray) and the number of filled components (black), bar plot of the relative frequencies of the number of filled components.
- Again three clusters are estimated, the final partition is the same as for $K = 3$ fixed.

Summary

- Mixtures are a very versatile model class for clustering and density estimation: mixtures of normals, of Bernoulli distributions (LCA models), of regression models, of Markov chains ...
- Bayesian mixtures has several advantages:
 - allows to regularize an overfitting mixture.
 - delivers a rich sampling output which allows to quantify model uncertainty (scatter plot of the draws)
- Next projects:
 - Biclustering (clustering of both variables and observations)
 - Clustering of matrix-variate data

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